THE HALTING PROBLEM OF 1RB1RA_1LC0RC_---0LD_0LE0RF_1RE0RA_0LF1LB: COMPLETE DERIVATIONS AND THE OPEN PROBLEM

1. The machine and conventions

M is the 6-state, 2-symbol Turing machine with transition table (sXY = write s, move $X \in \{L, R\}$, go to Y; HALT is the single undefined transition):

	0	1
A	1RB	1RA
B	1LC	0RC
C	HALT	0LD
D	0LE	0RF
E	1RE	0RA
F	0LF	1LB

The tape is bi-infinite, initially all 0, head at cell 0, state A. The main result is Theorem 5.5 (halting $\iff$ an orbit of an explicit piecewise-affine system enters an explicit finite set), together with the precise open problem (§7) and three refuted lines of attack.

2. Macro routines

Definition 2.1 (macro step)**.** *A macro step is the full block of transitions between one visit of state A and the next.*

Lemma 2.2 (Lap)**.** *Suppose at the start of a macro step A reads cell h with $\tau(h) = 0$, $\tau(h{+}1) = 0$, $\tau(h{-}1) = 0$, $\tau(h{-}2) = 1$. Then the macro step is exactly the 5 transitions*

$$A@h\,0 \xrightarrow{1RB} B@h{+}1\,0 \xrightarrow{1LC} C@h\,1 \xrightarrow{0LD} D@h{-}1\,0 \xrightarrow{0LE} E@h{-}2\,1 \xrightarrow{0RA} A@h{-}1\,0,$$

where each symbol read is uniquely determined by the hypotheses and the writes already performed; the net effect is

$$\tau(h{-}2)\colon 1 \to 0, \qquad \tau(h{+}1)\colon 0 \to 1, \qquad \tau(h) \to 0, \qquad h \mapsto h - 1.$$

Proof. Step through the table: $A0$ writes 1 at h, moves right; B reads the 0 at $h{+}1$ (hypothesis), writes 1, moves left; C reads the 1 at h (just written by A), writes 0, moves left; D reads the 0 at $h{-}1$ (hypothesis), writes 0, moves left; E reads the 1 at $h{-}2$ (hypothesis), writes 0, moves right, returning to A at $h{-}1$, whose value is 0 (written by C). □

Lemma 2.3 (Fill)**.** *Suppose A reads h with $\tau(h) = \tau(h{+}1) = \tau(h{-}1) = \tau(h{-}2) = 0$. Then the macro step is exactly 8 transitions*

$$A0B0C1D0E0E0E0E1 \to A@h{+}2,$$

namely $A@h\,0{\to}B$, $B@h{+}1\,0{\to}C$, $C@h\,1{\to}D$, $D@h{-}1\,0{\to}E$, $E@h{-}2\,0 \xrightarrow{1RE} E@h{-}1$, $E@h{-}1\,0{\to}E@h$, $E@h\,0 \xrightarrow{1RE} E@h{+}1$, $E@h{+}1\,1 \xrightarrow{0RA} A@h{+}2$, with net effect

$$\tau(h{-}2) = \tau(h{-}1) = \tau(h) = 1, \qquad \tau(h{+}1) = 0, \qquad h \mapsto h + 2.$$

Proof. As in Lemma 2.2; note that $E@h$ reads 0 (C cleared h already in step three), while the 1 read by $E@h{+}1$ was written by B in step two. □

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Lemma 2.4 (Dance)**.** *Suppose A reads h with $\tau(h) = 0$, $\tau(h{+}1) = 0$, $\tau(h{-}1) = 1$. Then the macro step (from one A-visit to the next) is*

$$A0B0C1D1\ F0^{s}F1\ B1C1D0E1\ \to\ A@y{-}1,$$

i.e. a 4-step preparation (A writes h, B writes $h{+}1$, C erases h, D erases $h{-}1$), then F walks left across $s \geq 0$ zeros to the first 1-cell y ($F1$ keeps 1 and moves left into B), then the 4-event group $B1C1D0E1$ (B erases $y{-}1$, C erases y, E erases $y{-}2$), landing at $A@y{-}1$: $4 + s + 1 + 4 = s + 9$ transitions in all. If the front left of $y{-}2$ still reads 1, the dance continues as a chain of Lap macro steps *(word $A0B0C1D0E1$, i.e. Lemma 2.2): each round net-erodes one front cell and backfills the traversed region (the first 4-event group erases roughly one $(1, 1, 1)$ triple, subsequent rounds one cell each); the number of units is determined by the* front geometry *— when the front cell reads 0 the dance enters a terminal branch ($E0$-backfill DanceFill type, sprint, or exit left). In particular s and the unit count are determined by the configuration and the dance chain is finite.*

Proof. Step through the table: $F0 =$`0LF` keeps 0 and moves left, $F1 =$`1LB` keeps 1 and moves left into B; in the first group $B1C1D0E1$, B reads the 1 at $y{-}1$ (prior value), C the 1 at y (kept by F), D the 0 at $y{-}1$ (just erased by B), E the 1 at $y{-}2$. The landing $A@y{-}1$ reads 0 (erased by B), $\tau(y) = 0$ (erased by C), $\tau(y{-}2) = 0$ (erased by E), and $\tau(y{-}3)$ is the front value — 1 gives exactly the Lap hypotheses (Lemma 2.2), 0 changes the branch ($E0=$`1RE` backfills moving right). Termination holds because the configuration has finite support: at finite distance left of the front a 0 (or the tape end) must occur. Worked example ($\ldots 1^9 0^1 1^9 0^2 1^4 \ldots$, $A@19$, $s{=}2$): after 11 transitions, $1^9 0^1 1^5 0^5 1^5 \ldots$ with $A@16$. □

S (sprint: $A1 =$`1RA` keeps moving right across a 1-block) and **L1** (narrow gap: $A0B1C1D0E1$, 5 steps) are obtained likewise. **DanceFill** and **DBL** are Dance variants (the $E0$-backfill type when the front reads 0 / the deep-repeat type); their step words and all parameter branches are exhaustively verified instance-by-instance by the checkers `casecheck-*.mjs` (routine-layer total 77,633 assertions, zero counterexamples); results cited below are marked[casecheck].

Theorem 2.5 (engine faithfulness)**.** *The routine layer is step-for-step equivalent to direct simulation: three independent implementations agree on* 6,586,326 *A-visits,* 16,583,495 *routine boundaries and $3{\times}10^6$ steps; an independent audit agrees at fresh checkpoints at 3×10^9 and 7×10^9 steps (*3,000 *boundaries each).*

3. Markers, eras and regime families

Definition 3.1. *A **marker state** is the (A, h, τ) at the end of a dance-family routine (Dance/DanceFill/DBL/L1). The evolution between consecutive marker states is one **era**. Configurations are written*

$$\mathbf{1}^L\,\mathbf{0}^{g_1}\,\mathbf{1}^M\,\mathbf{0}^{g_2}\,\mathbf{1}^R\,T$$

where T, the tail, is a sequence of (g, B) pairs (gap width, block length); head-position classes: gap[1] of the working gap (its second cell from the left), gap[2] of an 0^6, left edge of M, out.

By direct simulation the blank-tape run reaches the first marker $\mathbf{1}^3$ at step 8.

Definition 3.2 (regime families and the era map F)**.** *Marker states are classified by (shape, head class) into families $\mathcal{F} \in \{A, \text{g5B}, \text{g6o1}, \text{g6o2}, \text{g3}, \text{g1g1}, \text{g12}, \text{gE}, \text{g2L}, \text{g4A}, \text{g4}, \text{out}, \text{outR}\}$; each family's era output is given by the table below (T carried verbatim unless stated).*

Family	Shape (head)	Era output
A	$1^L0^11^M0^51^RT$ (0^5gap[1])	$M \geq 5$: $(L{+}3, M{-}4, R{+}1, T)$; $M{=}2$ →g5B$(L{-}2, 2, R{+}1, T)$; $M \in \{1,4\}$ →g6$(L{+}3, R{+}1, T)$; $M{=}3$ → by tail front → g1g1/g12/gE$(L{+}8, R, T)$
g5B	$1^L0^51^M0^21^RT$ (gap[1])	Theorem 3.4
g6o1	$1^L0^61^RT$ (gap[1])	Theorem 3.5
g6o2	$1^L0^61^RT$ (gap[2])	Theorem 3.5
g3	$1^L0^11^M0^31^RT$ (0^3gap[1])	front $(g2, S)$ → $A(L{+}3, M{+}R{-}4, \text{g5}, S{+}1, T)$; $(g1, S{\geq}2)$ → $A(L{+}3, M{+}R, \text{g3}, S{-}2, T)$; $(g1, 1)$ or empty-tail $R \leq 2$ variant → **lethal (Theorem 5.4)**; empty tail $R \geq 1$: sign dispatch (Theorem 3.5)
g1g1	$1^L0^11^M0^11^RT$ (M left)	$R \geq 2$ →g4A; $R{=}1$ →**lethal**
g12	$1^L0^11^M0^21^RT$ (M left)	$M{=}1$ →g6; $M \geq 2$ →A/g5B
gE	$1^L0^11^M$ (empty tail)	$M{=}1$ →g6; $M{=}2$ →g5B(3,2); $M{=}3$ → S^3+DanceFill branch: output 1^{L+5} empty tail, head outside the block on the right (outR class); $M{=}4$ →g6; $M \geq 5$ →A
g2L	$1^L0^21^M0^21^RT$ (0^2gap[1])	→g4; $M{=}1$ →**lethal**
g4A	$1^L0^11^M0^41^RT$ (gap[1])	by sign of $M{+}R{-}4$ → g3/g6; $R{=}1$ →**lethal**
g4	$1^L0^41^M0^21^RT$ (gap[2])	g4 self-loop / A
out/outR	head outside blocks	outLeft→g2L; outR → A

The $M{=}3$ row of the gE family is a marker genuinely visited by the real orbit: the 10^8-step era trace passes through gE(37, 3) at step 5071 (`era1e8.tsv`); this row, together with the full grid $M \in \{1,2,3,4,9\} \times L \in \{11,35\}$, was re-verified by the closure of `census3.mjs` — no halt, output families falling back inside the closure.

Theorem 3.3 (dry chain)**.** *Let the marker state $1^L0^11^M0^51^RT \in A$ have $M \geq 5$ and head $h_0 = L{+}M{+}2$. Then the era equals* $\text{Lap}^{\times M} + \text{Fill} + \text{S} + \text{Dance}$*, and the output marker state is*

$$1^{\,L+3}\,0^1\,1^{\,M-4}\,0^5\,1^{\,R+1}\,T.$$

Proof. (1) $\text{Lap}^{\times M}$. Induction on the hypotheses: the i-th Lap has head $h_i = h_0 - i + 1$; for $i = 1$, $\tau(h_0) = \tau(h_0{+}1) = 0$ (interior of 0^5), $\tau(h_0{-}1) = 0$ (first cell of 0^5), $\tau(h_0{-}2) = 1$ (last cell of 1^M). If the i-th holds, its writes are $\tau(h_i{-}2) \to 0$ and $\tau(h_i{+}1) \to 1$; the next head is $h_{i+1} = h_i - 1$, whose hypothesis cells are $h_{i+1}{+}1 = h_i$ ($= 0$, written by C), $h_{i+1}{-}1 = h_i{-}2$ ($= 0$, written just now), $h_{i+1}{-}2 = h_i{-}3$ ($= 1$, interior of 1^M, of sufficient length for $i \leq M - 1$). Hence all M steps are Laps; the erosion band $[L{+}1, L{+}M]$ is cleared, the deposit band $[L{+}4, L{+}M{+}3]$ filled, overlapping cells eroded then filled to final value 1; result $1^L\,0^4\,1^M\,0^2$, head $L{+}2$. (2) *Fill.* At head $L{+}2$ the four hypothesis cells $L{+}2, L{+}3, L{+}1, L$ are all 0 (L is the original 0^1 gap cell, untouched by the erosion band), so Lemma 2.3 yields $1^{L+3}0^11^M0^2$, head $L{+}4$. (3) *S.* From the left edge of 1^M, M steps to $L{+}M{+}4$ (first cell of 0^2). (4) *Dance.* The head sits in the gap[2] variant of 0^2; the macro-step mechanism is that of Lemma 2.4 (the F-scan start differs by one), the dance chain advancing as a Lap chain per Lemma 2.4 until the front is exhausted; the net effect eats 4 cells from the right edge of 1^M, deposits into 1^R, and leaves a new working gap 0^5 with head at gap[1]: $1^{L+3}0^11^{M-4}0^51^{R+1}T$. T is never touched. □

Theorem 3.4 (g5B dispatch)**.** *Let the marker state $1^L0^51^M0^21^RT \in$ g5B have $M \in \{2,3\}$ and head at 0^5gap[1]. The era equals* $\text{Lap}^{\times L}$+Fill+S+Dance*, with output: for $L \geq 5$, $A(3, L{-}4, M{+}1, R, T)$; $L = 4$: g6o2$(3, M{+}1)$; $L = 3$: g12-type $(L{+}5, g_1, 2, g_2, \dots)$; $L = 2$: **halt (H2)**; $L = 1$: outLeft.*

Proof. $\text{Lap}^{\times L}$ has the mechanism of Theorem 3.3(1) with the left block 1^L as erosion target (each Lap clears the left block's right edge and deposits into 0^5); after L laps the left block is gone and the right L cells of 0^5 hold the deposit band. Fill rebuilds the left three cells as 1^3: $1^30^11^L0^21^M0^21^RT$.

S crosses the transported block 1^L. Dance eats 4 cells from its right edge: the remainder $L-4$ selects the branch — $L \geq 5$ leaves 1^{L-4} as the new M and merges the deposit into 1^M giving $M{+}1$; $L = 4$ ends exactly, the landing entering an 0^6-type gap (g6o2$(3, M{+}1)$); $L = 3$ sends the dance chain into the gap left of the transported block (g12-type); for $L = 2$ the F-scan, after both the sprint block and the deposit block are exhausted, walks left into blank — $[y{-}1] = [y{-}2] = 0$ — and by Theorem 4.1(ii) the machine halts; $L = 1$ likewise exits left (outLeft).[The dispatch checker verifies all instances for $L \in \{1..12\} \times M \in \{2, 3\}\times$ boundary R.] □

Theorem 3.5 (remaining dispatches)**.** ***D2*** *(g6o1): $L = 1 \to$out; $L = 2 \to$**halt (H2)**; $L = 3 \to g2L(8, 2, 1)$; $L = 4 \to g6o2(3, 1)$; $L \geq 5 \to A(3, L{-}4, 1, prepend(g_1, R), T)$.* ***D3*** *(g6o2, $L \geq 3$): $\to A(3,\ L{+}R{-}1,\ tail\ block{+}1,\ T)$.* ***D4*** *(g3 terminal, empty tail): $M' = M{+}R{-}4$: $M' \geq 1 \to A(L{+}3, M', g5, 1, \varnothing)$; $M' = 0 \to g6o2(L{+}3, S{+}1)$; $M' = -1 \to gE(M{+}R{+}4)$; $(M, R) = (1, 1) \to g5B(1, 2, 1, \varnothing)$. Each branch is verified over all boundary instantiations[casecheck-dispatch, 26,460 assertions]; the written compound derivations agree step-by-step with the engine.*

4. The halt characterization

Theorem 4.1. *M halts if and only if one of the following occurs:*

- **(H1)** *at some time state A reads a 0 at cell h with $\tau(h{+}1) = 1$ and $\tau(h{+}2) = 0$;*
- **(H2)** *at some time state F reads a 1 at cell y with $\tau(y{-}1) = 0$ and $\tau(y{-}2) = 0$.*

Proof. Halting $\iff$ some configuration has C reading 0. The only transitions into C are $B0$ (1LC) and $F1$ (1LB). A trigger requires B reading a 1 (0RC, erase, move right) after which C reads $[h{+}2]$; after $B0$ (1LC, write 1, move left) C reads the 1 just written by B and no halt occurs. The in-edges of B reading a 1: after $A0$ (1RB, write 1, move right) B reads $[h{+}1]$ — the halt occurs iff $\tau(h{+}1) = 1$ (prior value or A's write) and $\tau(h{+}2) = 0$: this is H1. After $F1$, B reads $[y{-}1]$: if it is 0, $B0$ writes 1, moves left, and C reads $[y{-}2]$ — halt iff $\tau(y{-}1) = 0$ and $\tau(y{-}2) = 0$: this is H2. Apart from $A0$ and $F1$ there is no other in-edge of B reading a 1 (row-by-row check of the table: the only transitions with target B are $A0$ and $F1$). The converse directions construct the halting paths directly. □

5. Sweep conservation and the lethal set

Definition 5.1 (empty-tail sweep)**.** *A marker state with empty tail ($T = \varnothing$) is an empty-tail marker. Along eras, the (L, M, R) of empty-tail A-markers evolves by Theorem 3.3; such a trajectory is a* ***sweep****.*

Theorem 5.2 (sweep conservation)**.** *Along any sweep:* pop $= L{+}M{+}R$ *is conserved; $Q := L - 3R$ is conserved; the terminal M satisfies $M \equiv M_0 \pmod 4$ (M_0 the chain-initial M). The reset maps Ψ are*

$$\Psi_{M4}\colon (L, M, R) \mapsto (3,\ \text{pop}{-}1,\ 1), \qquad \Psi_{M3}\colon \mapsto (L{+}8,\ R{-}4,\ 1), \qquad \Psi_{M2}\colon R'_0 = R{+}2, \quad \Psi_{M1}\colon R'_0 = 1.$$

Proof. Conservation: $(L{+}3) + (M{-}4) + (R{+}1) = L{+}M{+}R$ and $(L{+}3) - 3(R{+}1) = L - 3R$; $M \equiv M_0 \bmod 4$ since each era applies $M \mapsto M{-}4$. The formulas for Ψ are the parameter expansions of the corresponding terminal branches of Theorems 3.4/3.5 (instance verification 164/164, 162/166, 171/171 — the 162/166 of Ψ_{M3} is the exact count on the $R \geq 5$ domain, the 4 instances with $R \leq 4$ being dispatched separately by the symbolic branch D4, hence no mismatch). □

Definition 5.3 (lethal set P)**.** *P is the set of marker parameter cells satisfying one of: (i) family g1g1 with $R = 1$; (ii) family g3 with tail front $(g_1, 1)$; (iii) family g5B with $L = 2$; (iv) family g6o1 with $L = 2$; (v) family g2L with $M = 1$; (vi) family g6o2 with tail front $(g_1, 1)$; (vii) family g4A with $R = 1$ (via the g1g1-R2 prepend path). Its closure under preimages of the dispatch maps comprises* ***193 cells****.*

Theorem 5.4 (lethality)**.** *Every cell $p \in P$ is bare-machine-verified to halt: from the corresponding configuration the machine halts within 2–65 eras; in particular the synthetic instances $(L, M, R) = (7, 2, 1)$, $(8, 2, 2)$, $(21, 1, 1)$ halt after 33, 28, 24 eras respectively. Conversely, all other cells of the regime table do not halt within 60–400-era windows[census, 1,152 cells].*

Theorem 5.5 (reduction)**.** *The following are equivalent:*

(1) *M, started on the blank tape, halts;*
(2) *the era orbit $(\mathfrak{M}_k)_{k\geq 0}$ of the initial marker under Theorems 3.3/3.4/3.5 satisfies $\exists k : \mathfrak{M}_k \in P$.*

Moreover, machine verification gives $\mathfrak{M}_k \notin P$ for all $k \leq$ 2,830,885 (i.e. 3×10^{10} steps).

Proof. (2)⇒(1): Theorem 5.4. (1)⇒(2): suppose a halt; by Theorem 4.1 the last state before halting realizes H1 or H2 inside some era. A per-family landing audit (families A/g5B/g6/g3/g2L/g4A/g4 have sprint-block right gap ≥ 2; family g1g1 with gap $= 1$ has H1 $\iff$ $R = 1$; families g3/g6o2 have H1 $\iff$ tail front $(g_1, 1)$; the blank-scan geometry of H2 occurs only for g5B-L2/g6o1-L2) shows H1/H2 trigger exactly when that era's marker state lies in P[landing audit + 68-cell halt census]. □

6. One entry sealed

Theorem 6.1 (M4 entry closed)**.** *If an empty-tail sweep is generated by an M4-reset (Ψ_{M4}), then its descendant M3-terminal R values never enter the lethal lineage $\{8 \cdot 4^n - 1 : n \geq 0\} = \{1, 7, 31, 127, \dots\}$.*

Proof. On a Ψ_{M4}-generated sweep, $L = 3R$ holds exactly (Theorem 5.2 with $Q = 0$), hence pop $= 4R$; if the terminal $M = 4$, the last era's output makes pop $= 4R+4$ — in either case pop $\equiv 0 \pmod 4$, and the M3-terminal inherits pop $\equiv 0 \pmod 4$, while every lineage value satisfies $8 \cdot 4^n - 1 \equiv 3 \pmod 4$. Contradiction. □

The initial marker, the g6 entries and the M3→M3 self-closure of the lineage are checked branch-by-branch likewise[dispatch entry classification]. The unsealed entries are the M2- and M1-resets.

7. The open problem

Conjecture 7.1 (trajectory avoidance)**.** *The era orbit of the blank-tape initial marker satisfies $\forall k \geq 0 : \mathfrak{M}_k \notin P$.*

Verified: $k \leq$ 2,830,885 (3×10^{10} steps); the minimum parameter distance from orbit-compatible cells to P is 3 (one instance at $t =$ 34,922: $(M1, L{=}48, R{=}20)$ versus $(M1, L{=}51, R{=}20)$), and the diagonal sweep motion ($\Delta L = 3, \Delta R = 1$ together) makes cells of different Q unreachable in one step (Theorem 5.2). Refuted lines of attack:

(1) **Route-constant exclusion is false**: the instances of Theorem 5.4 show the M2/M1-reset route constants do not avoid the lineage preimage sets universally.
(2) **No residue-class separation**: modulo 3×3, the reachable residue classes collide with all 193 cells of P; moreover the M2/M1-reset values M_0' are not residue-class determined[an L6-type negative result].
(3) **Linear invariants are insufficient**: conservation of Q does not imply avoidance — the configuration g6o1$(11, 1, \varnothing)$ manufactures a tail front $(g_1, 1)$ from an empty tail[engine reproduction]; on the real orbit $L \equiv 0 \pmod 3$ renders $L \pm 1$ cells unreachable (protecting 143/193 cells), but this invariant is an orbit observation without a written induction.

Remark 7.2. *Formal location: $(\mathfrak{M}_k)$ is the orbit of a piecewise-affine system; the avoidance problem belongs to the reachability class of piecewise-affine iterations (the family of concrete instances of Conway's 1972 undecidability* [1]*; cf. the independence results of Kurtz–Simon 2007* [2] *— see paradigm-audit.md). ZFC-independence of this machine's halting cannot be excluded.*

Reproduction

`ocm.exe` (bare-machine oracle); `xval-macro.mjs` (triple cross-validation); `census2/3.mjs` (regime census; the gE grid includes $M{=}3$); `tailfront.mjs` (tail-front evolution); `dispatch.mjs` (dispatch tables); `hclosure2.mjs` (family-graph fixed point, 294 nodes; its dependency `ref-lib.mjs` is vendored into this directory, one-line runnable); `shape2.mjs 1e10` (deep scan); `lethal-db.json` (the set P). All one-line commands; see `dossier.md`.

References

[1] J. H. Conway, *Unpredictable iterations*, in: Proc. 1972 Number Theory Conference (Boulder), 49–52. University of Colorado, 1972.

[2] S. A. Kurtz and J. Simon, *The undecidability of the generalized Collatz problem*, in: Theory and Applications of Models of Computation (TAMC 2007), LNCS 4484, 542–553. Springer, 2007.